

Progress Report: Sprint 8

CS.028 Optimizing Research Software Codes - 2 February 2026

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Summary

- Focused on resolving xarray.merge conflicts, still some bugs present.
- Refactored README.md, CONTRIBUTING.md, and organized directory.
- Beginning code segmentation to distribute for teammates to work on.

Sprint Work Log

Working on Xarray Merge Error Using compat

Status: Ongoing (~70% complete)

Author: Kathryn Butler

Reviewers: N/A

Source Link: [Issue 7](#)

Supporting Artifact: [xarray.merge](#)

[Merge multiple xarray Datasets with overlapping coordinates - Stack Overflow](#)

Research done to understand what xarray.merge is responsible for

```
PS C:\Projects\optimizing-research-software-codes> cd .\gnss_python-main\  
PS C:\Projects\optimizing-research-software-codes\gnss_python-main> python rnx2db.py
```

Lack of xarray.merge error when run with {compat='override'}

Notes: Due to this bug fix, another has appeared:

```
C:\Projects\optimizing-research-software-codes\gnss_python-main\georinex\obs2.py:54: FutureWarning: In a future  
version of xarray the default value for join will change from join='outer' to join='exact'. This change will r  
esult in the following ValueError: cannot be aligned with join='exact' because index/labels/sizes are not equal  
along these coordinates (dimensions): 'sv' ('sv',) The recommendation is to set join explicitly for this case.  
obs = xarray.merge((obs, o), compat='override')
```

New error message

It's related to the join= 'exact' versus join='outer', which is another issue to address. Additionally, setting compat='override' interfered with the output, which means I need to reassess how the program handles merges, and the correct argument to set compat to.

README and Directory Organization

Status: Complete

Author: Kathryn Butler

Reviewers: N/A

Source Link: [README.md](#), [CONTRIBUTING.md](#), [updated directory structure](#)

Supporting Artifact:

```
▼ README.md
19 19 ---
20 20 @@ -22,15 +22,18 @@ Specifically, the inherited RINEX2DB Python program converts RINEX observation a
21 21
22 22
23 23 | Directory | Description |
24 24 |-----|-----|
25 25 - | `datasets/COVL_1sec/` | Sample input datasets (observation and navigation files). |
26 26 + | `.github/workflows/` | Code analysis folder (currently disabled). |
27 27 + | `Datasets/COVL_1sec/` | Sample input datasets (observation and navigation files). |
28 28 + | `Testing/` | Testing functions and data go here (currently empty). |
29 29 + | `documentation/` | Architecture, contributing, and progress reports. |
30 30
31 31 | `gnss_python-main/` | Main program folder containing `rnx2db.py` and supporting scripts. |
32 32 - | `doc/` | Project documentation (setup, usage, and background). |
33 33 - | `.gitignore` | Python `.gitignore` file. |
34 34
35 35 ---
36 36
37 37 ## Setup & Installation
38 38
39 39 - Instructions for downloading and running the project will be added after testing is complete.
40 40
41 41 + [Current Setup and Run Instructions](https://github.com/KathrynJButler/optimizing-research-software-
42 42 codes/blob/main/documentation/How_to_Run.pdf)
43 43 +
44 44 + This process will be updated after further progress is made to reduce runtime.
```

Sample of the README.md [commit](#) history and directory updates

Notes: Updated the README to clarify setup instructions and project structure. Reorganized the directory layout to remove unnecessary files, separated documentation, and allowed for better GitHub file navigation.

Individual Code Adjustments

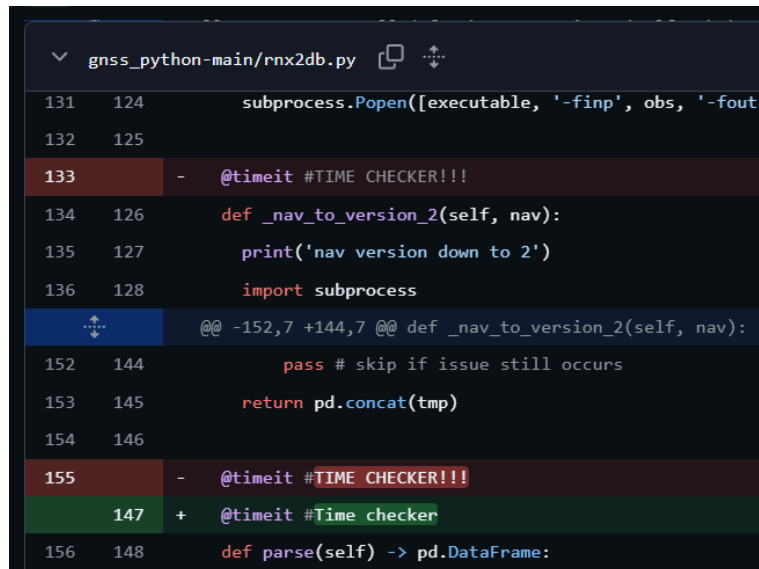
Status: Ongoing (~30% complete)

Author: Kathryn Butler

Reviewers: N/A

Source Link: [Issue 13](#) and [Issue 4](#)

Supporting Artifact:



```
gnss_python-main/rnx2db.py
131 124 subprocess.Popen([executable, '-finp', obs, '-fout'
132 125
133 - @timeit #TIME CHECKER!!!
134 126 def _nav_to_version_2(self, nav):
135 127     print('nav version down to 2')
136 128     import subprocess
@@ -152,7 +144,7 @@ def _nav_to_version_2(self, nav):
152 144     pass # skip if issue still occurs
153 145     return pd.concat(tmp)
154 146
155 - @timeit #TIME CHECKER!!!
147 + @timeit #Time checker
156 148 def parse(self) -> pd.DataFrame:
```

Section of [commit](#) to only time check `parse()` and `calculate_positions()`

Notes: The current `parse()` and `calculate_positions()` functions are confusing to track and optimize as full functions. The time check function now only tracks the problematic functions, which means it will interfere less with the total runtime. I am currently working on finding all the functions which `parse()` and `calculate_positions()` depend on, and then assign them to each teammate to focus on optimizing independently.

Risk and Quality Tracking

Bug Count: 6 ([Link to issue query](#))

Top Bugs:

- [Treating Keys as Positions and Using Argsort](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Series.getitem treating keys as positions is deprecated. Look up how to solve it.)
- [Incorrect conversion from float to datetime](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved

- **Actions:** No further action needed. (Research into the line causing the problem. Determine the input causing the problem, and correct the datetime variable.)
- [Log.txt hardcoded](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Resolved
 - **Actions:** No further action needed. (Log.txt seems to be announcing a preset number of cores to calculate. Figure out why it is set to only be 8 cores. Find out if you can manually adjust the core count to be something other than just 8, then add the text to the function dynamically.)
- [Inconsistent timing data](#)
 - **Severity:** High
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Create a smaller test data pool, run the tests multiple times to determine if the size of the file increases the likelihood of failed parsing attempts.
- [Xarray.merge conflict](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Set compat explicitly to something other than 'override', then test output for clarity. (Previous goal: Research what "compat" does, and either set it explicitly or use the new combine_kwargs_defaults.)
- [Numpy DataFrame.swapaxes is deprecated](#)
 - **Severity:** Medium
 - **Owner:** Kathryn Butler
 - **Current Status:** Ongoing
 - **Actions:** Research into the numpy process and determine if this bug affects runtime. If so, attempt to update the numpy version. Additionally, research into how using the most up-to-date libraries affect runtime/output consistency.

Next Sprint Plan

- Resolve xarray.merge conflict
 - Investigate and resolve the merge conflict without relying on `compat='override'`. Compare behavior of `join='exact'` vs `join='outer'` using controlled test cases. Source of truth: [Issue 11](#) and [Issue 7](#)
- Create reproducible testing data from existing dataset
 - Generate a smaller dataset to reproduce runtime inconsistencies. Establish baseline timing benchmarks and document results. Commit dataset to repository. Source of truth: [Issue 4](#)
- Split code optimization into modular sections
 - Break the current problem functions (`parse()`, `calculate_positions()`) into modules, and assign ownership to specific team members. Update README with new structure and responsibilities. Source of truth:

Team Process Reflection

This sprint was more stressful than expected because one teammate did not complete any assigned work and had no communication with the rest of the team. This required the remaining members to take on additional responsibilities to keep progress moving. We addressed the issue by speaking with the TA, our project partner, and Professor Winters to make sure expectations were clear and documented moving forward.

Despite the challenges, we were able to make progress on debugging merge conflicts and improving documentation. We've discovered we need stronger task ownership and clearer accountability within the team. Moving forward, we plan to divide work more explicitly and set clearer check-ins to prevent similar issues.